Decimal to Binary Conversion

Represent the following numbers in binary — Answer:

512 → 1000000000_2

44 → 101100_2

129 → 10000001_2

717 → 1011001101_2

999 → 1111100111_2

Name in Binary

Write your name in binary (Mustafa) — Answer:

M → 01001101

u → 01110101

s → 01110011

t → 01110100

a → 01100001

f → 01100110

a → 01100001

Mustafa →

01001101 01110101 01110011 01110100 01100001 01100110 01100001

Hexadecimal to Decimal Conversion

Convert the following hexadecimal numbers to decimal — Answer:

$12A4_{16}$ → 4772_{10}

$1C35_{16}$ → 7221_{10}

100_{16} → 256_{10}

$115C_{16}$ → 4444_{10}

Decimal to Hexadecimal Conversion

Convert the following decimal numbers to hexadecimal — Answer:

$4722_{10} \rightarrow 1272_{16}$

$7221_{10} \rightarrow 1C35_{16}$

$256_{10} \rightarrow 100_{16}$

$4444_{10} \rightarrow 115C_{16}$

Hexadecimal to Binary Conversion

Convert the following hexadecimal numbers to binary — Answer:

$13B4_{16} \rightarrow 0001\ 0011\ 1011\ 0100_{2}$

$A5_{16} \rightarrow 1010\ 0101_{2}$

Basic Digital Data Units — Answers

1. What is bit?

A **bit (binary digit)** is the smallest unit of data in a computer. It can have a value of **0 or 1**.

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2. What is Crumb?

A **crumb** is a group of **2 bits**.

---

3. What is Nibble?

A **nibble** is a group of **4 bits**.

---

4. What is Byte?

A **byte** is a group of **8 bits** used to represent data or characters.

---

5. How many bits in a byte?

8 bits

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6. How many crumbs in a byte?

4 crumbs (because 1 crumb = 2 bits, and 8 ÷ 2 = 4)

---

7. How many nibbles in a byte?

2 nibbles (because 1 nibble = 4 bits, and 8 ÷ 4 = 2)

---

8. How many crumbs in a nibble?

2 crumbs (because 1 nibble = 4 bits, and 4 ÷ 2 = 2)

---

9. What is the lower nibble?

The **lower nibble** is the **rightmost 4 bits** of a byte.

---

10. What is the upper nibble?

The **upper nibble** is the **leftmost 4 bits** of a byte.

---

11. What is LSB? And why we call it LSB? What is its value?

LSB (Least Significant Bit) is the **rightmost bit** in a binary number.

It is called LSB because it has the **smallest value ($2^0 = 1$)**.

---

12. What is MSB? And why we call it MSB? What is its value?

MSB (Most Significant Bit) is the **leftmost bit** in a binary number.

It is called MSB because it has the **largest value** depending on the number of bits (in an 8-bit number its value is **$2^7 = 128$**).

Hexadecimal to Binary Conversion (Direct)

Convert the following hexadecimal numbers to binary — Answer:

$12A4_{16} \rightarrow 0001\ 0010\ 1010\ 0100_2$

$1C35_{16} \rightarrow 0001\ 1100\ 0011\ 0101_2$

$100_{16} \rightarrow 0001\ 0000\ 0000_2$

$115C_{16} \rightarrow 0001\ 0001\ 0101\ 1100_2$

Binary to Hexadecimal Conversion (Direct)

Convert the following binary numbers to hexadecimal — Answer:

$0001\ 0010\ 1010\ 0100_2 \rightarrow 12A4_{16}$

$0001\ 1100\ 0011\ 0101_2 \rightarrow 1C35_{16}$

$0001\ 0000\ 0000_2 \rightarrow 100_{16}$

$0001\ 0001\ 0101\ 1100_2 \rightarrow 115C_{16}$

Octal to Decimal Conversion

Convert the following octal numbers to decimal — Answer:

$100_8 \rightarrow 64_{10}$

$512_8 \rightarrow 330_{10}$

Decimal to Octal Conversion

Convert the following decimal numbers to octal — Answer:

$64_{10} \rightarrow 100_8$

$330_{10} \rightarrow 512_8$

Octal to Binary Conversion (Direct)

Convert the following octal numbers to binary — Answer:

$100_8 \rightarrow 001\ 000\ 000_2$ (or 1000000_2)

$512_8 \rightarrow 101\ 001\ 010_2$ (or 101001010_2)

Binary to Octal Conversion (Direct)

Convert the following binary numbers to octal — Answer:

$0100\ 0000_2 \rightarrow 100_8$

$0001\ 0100\ 1010_2 \rightarrow 1242_8$

Computer Networks — Answers

- **What is a network?**

A **network** is a group of computers and devices connected together to share data, resources, and information.

- **Why do we need a network?**

We need networks to:

- Share files and data
- Share hardware (printers, internet)
- Communicate easily (email, messages)
- Access information quickly

- **What is LAN vs WAN?**
- **LAN (Local Area Network):** A network that covers a small area like a home, school, or office.
- **WAN (Wide Area Network):** A network that covers a large area like cities or countries (example: the Internet).

- **How do computers communicate together?**

Computers communicate by sending and receiving data through cables or wireless signals using rules called **protocols**.

- **What is Ethernet?**

Ethernet is a wired networking technology that connects computers using cables to share data.

- **What is Wireless?**

Wireless communication connects devices without cables using radio signals.

- **What is Protocol?**

A **protocol** is a set of rules that control how data is sent and received between devices in a network.

- **What is Wi-Fi?**

Wi-Fi is a wireless technology that allows devices to connect to the internet or a network without cables.

- **Wi-Fi stands for what?**

Wi-Fi does not officially stand for anything — it is a brand name (not “Wireless Fidelity,” which is a common misconception).

Programming Basics — Answers

- **What is a Programming Language?**

A **programming language** is a language used to write instructions that tell a computer what to do.

- **What is Code?**

Code is a set of instructions written in a programming language to perform a specific task.

- **What is Source Code?**

Source code is the original program written by a programmer in a programming language that humans can read.

- **What is Object Code?**

Object code is the machine-readable code produced after compiling source code, which the computer can execute.

- **Why do we need Compilers or Interpreters?**

We need **compilers or interpreters** to translate source code into machine language so the computer can understand and run it.

- **When is the language considered Fast?**

A language is considered **fast** when it executes programs quickly and is close to machine language (low-level languages).

- **When is the language considered Slow?**

A language is considered **slow** when it requires more translation steps and is far from machine language (high-level languages).

- **What is a high-level language?**

A **high-level language** is a programming language that is easy for humans to read and write (close to human language).

- **What is a low-level language?**

A **low-level language** is a programming language close to machine language and harder for humans to understand.

- **When do we consider a language is human readable?**

A language is **human readable** when its syntax and instructions are easy for people to understand and similar to natural language.

Programming Translation Tools — Answers

- **What is a compiler? And how it works?**

A **compiler** is a program that translates the whole source code into machine code at once before execution. It checks errors and produces an executable file.

- **What is an Interpreter? And how it works?**

An **interpreter** translates and executes the source code line by line during program execution without creating an executable file.

- **What is an Assembler?**

An **assembler** converts assembly language (low-level language) into machine code.

- **What is a Linker?**

A **linker** combines object files and libraries together to create a final executable program.

- **What is an exe file?**

An **.exe file** is an executable file that contains machine code and can run directly on a computer.

- **What is a Loader?**

A **loader** is a program that loads the executable file into memory so the computer can run it.

- **Which is faster Compiled or Interpreted languages?**

Compiled languages are faster because the program is translated before execution.

- **Does Interpreter produce exe file?**

No, an interpreter does not produce an executable (.exe) file.

- **Does Interpreter save machine code?**

No, it translates and executes code without saving machine code.

- **Does Compiler produce exe file?**

Yes, a compiler produces an executable (.exe) file.

- **Does Compiler save machine code?**

Yes, the compiler generates and saves machine code.

Logical Expressions — Answers

First Column

$12 \geq 12$ → **True**

$12 > 7$ → **True**

$8 < 6$ → **False**

$8 = 8$ → **True**

$12 \leq 12$ → **True**

7 = 5 → **False**

Second Column

NOT (12 ≥ 12) → **False**

NOT (12 < 7) → **True**

NOT (8 < 6) → **True**

NOT (8 = 8) → **False**

NOT (12 ≤ 12) → **False**

NOT (7 = 5) → **True**

Third Column

1 AND 1 → **1 (True)**

True AND 0 → **0 (False)**

0 OR 1 → **1 (True)**

0 OR 0 → **0 (False)**

NOT 0 → **1 (True)**

NOT (1 OR 0) → **0 (False)**

Fourth Column

(7 = 7) AND (7 > 5) → **True**

(7 = 7) AND (7 < 5) → **False**

(7 = 7) OR (7 < 5) → **True**

(7 < 7) OR (7 > 5) → **True**

NOT (7 = 7) AND (7 > 5) → **False**

(7 = 7) AND NOT (7 < 5) → **True**

Arithmetic Expressions — Answers

Left Column

$6 \div 3 + 4 \times 2 \rightarrow$ **10**

$6 \times 4 - 12 \div 3 - 8 \rightarrow$ **12**

$4 \times 4 - 3 \times 3 - 16 \div 4 \rightarrow$ **3**

$20 - (3 \times 23 - 5) \rightarrow$ **−44**

$(5 + 2)^2 - 9 \times 3 + 2^3 \rightarrow$ **30**

$(7 - \sqrt{9}) \times (42 - 3 + 1) \rightarrow$ **160**

Right Column

$5\times(2^2 + 3) + (2^3)^2 \rightarrow$ **99**

$6\times2 + 5\times1 + 4/2 - 1 \rightarrow$ **18**

$1 + (2\times(4 - 3 + 1) + 7) - 2 \rightarrow$ **10**

$10 + 6 \times (3 - 2) \times 8 / 24 - 1 \rightarrow$ **11**

$(20 - 18)^3 / 8 \times 3 - 1 \rightarrow$ **2**

$(2 + 1 - 3 + 1 + 4)^2 - 2\times4 \rightarrow$ **17**

Logical Expressions — Answers

(5 > 6 and 7 = 7) OR (1 OR 0) → **True**

NOT(5 > 6 and 7 = 7) OR (1 OR 0) → **True**

NOT(5 > 6 and 7 = 7) OR NOT (1 OR 0) → **True**

NOT(5 > 6 OR 7 = 7) AND NOT (1 OR 0) → **False**

((5 > 6 AND 7 ≤ 8) OR (8 > 1 and 4 ≤ 3)) AND True → **False**

((5 > 6 AND NOT(7 ≤ 8)) AND (8 > 1 OR 4 ≤ 3)) OR True → **True**

Programming Variables — Answers

- **What is Variable? And why do we need it?**

A **variable** is a named storage location in memory used to store data that can change during program execution.

We need it to store and manipulate data in programs.

- **What is Constant? And why do we need it?**

A **constant** is a value that cannot be changed during program execution.

We need it to store fixed values like π or tax rates.

- **What is Memory Cell?**

A **memory cell** is a small storage location in computer memory used to store one piece of data.

- **What is Identifier?**

An **identifier** is the name given to a variable, function, or program element to identify it in a program.

- **What is Memory Address? What is its relation to location?**

A **memory address** is a unique number that identifies a memory location.

It represents the exact location of a memory cell in the computer's memory.

- **Which numbering system is used in memory address?**

Hexadecimal (base 16) numbering system.

- **Mention the primary types of variables?**
- Numeric (numbers)
- String (text)
- Boolean (true/false)

- **What are the number's types? And give examples?**
- **Integer:** whole numbers (example: 10, −5)
- **Float/Real:** decimal numbers (example: 3.14, 2.5)

- **What is string? And give an example on it.**

A **string** is a sequence of characters or text.

Example: `"Hello"` or `"Mustafa"`.

- **What is Boolean? And give an example on it.**

A **Boolean** is a data type that has only two values: **True** or **False**.

Example: `True`, `False`, or `5 > 3` (True).

Programming Variables & Data Types — Answers

- **Are all variables and constants have the same size?**

No, variables and constants can have different sizes depending on their data type (integer, float, string, etc.).

- **What happens when you use much larger size than you need in variables?**

It wastes memory and may reduce program efficiency.

- **Can we modify a variable during the program?**

Yes, a variable's value can be changed during program execution.

- **Can we modify a constant during the program?**

No, a constant's value cannot be changed after it is defined.

- **How to make a variable read only?**

By declaring it as a **constant** (using keywords like `const` or `final` depending on the programming language).

- **Where do the variables and constants get stored?**

They are stored in the computer's **memory (RAM)**.

- **What is the difference between Integers and Floats?**
- **Integer:** stores whole numbers (example: 5, −10).

- **Float:** stores decimal numbers (example: 3.14, 2.5).

Variables in Memory — Answers (Fill the ???)

For the statement: `int Age = 45`

- **int** → **Data Type** (Integer type of variable)
- **Age** → **Variable Name / Identifier**
- **=** → **Assignment Operator** (assigns value)
- **45** → **Value (Data stored)**

RAM Image (top) → **Main Memory (RAM)** — where variables are stored.

Square Box (45) → **Memory Cell** — stores the value.

- **Name:** Age
- **Value:** 45
- **Address:** Memory Address (unique location in memory)

Memory Location → The specific place in memory where the variable is stored.

Logical Expressions — Answers

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7 = 5 → **False**

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NOT (8 = 8) → **False**

NOT (12 ≤ 12) → **False**

NOT (7 = 5) → **True**

Third Column

1 AND 1 → **1 (True)**

True AND 0 → **0 (False)**

0 OR 1 → **1 (True)**

0 OR 0 → **0 (False)**

NOT 0 → **1 (True)**

NOT (1 OR 0) → **0 (False)**

Fourth Column

(7 = 7) AND (7 > 5) → **True**

(7 = 7) AND (7 < 5) → **False**

(7 = 7) OR (7 < 5) → **True**

(7 < 7) OR (7 > 5) → **True**

NOT (7 = 7) AND (7 > 5) → **False**

(7 = 7) AND NOT (7 < 5) → **True**